

Shareholder Cash Productivity and the Cross Section of Stock Returns

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October 21, 2025

Abstract

This paper studies the asset pricing implications of Shareholder Cash Productivity (SCP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on SCP achieves an annualized gross (net) Sharpe ratio of 0.63 (0.60), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 38 (37) bps/month with a t-statistic of 3.62 (3.53), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash-based operating profitability, Operating profitability RD adjusted, Amihud's illiquidity, gross profits / total assets, Return on assets (qtrly), Past trading volume) is 23 bps/month with a t-statistic of 2.21.

1 Introduction

Asset prices reflect firms’ production decisions and the frictions they face when transforming inputs into outputs. In competitive markets, cross-sectional variation in expected returns arises from heterogeneity in firms’ exposure to systematic productivity shocks and their ability to adjust production in response to changing economic conditions. Despite extensive research on production-based asset pricing models, we lack comprehensive evidence on how cash flow productivity—the efficiency with which firms convert operating activities into shareholder distributions—relates to expected returns in the cross-section. This paper fills this gap by examining Shareholder Cash Productivity (SCP), defined as the ratio of cash distributions to shareholders relative to operating cash flows, as a predictor of stock returns through the lens of production-based asset pricing theory.

In a production economy, firms with higher shareholder cash productivity face lower marginal costs of capital adjustment and experience smaller frictions when scaling production [Cochrane \(1991\)](#); [Zhang \(2005\)](#). These firms optimally choose higher investment rates when productivity shocks are positive, leading to procyclical cash flows that command higher risk premia. The q-theory of investment predicts that firms with efficient cash conversion mechanisms maintain lower adjustment costs, enabling more responsive capital allocation that amplifies their exposure to aggregate productivity risk [Liu et al. \(2009\)](#). This enhanced sensitivity to systematic shocks manifests as higher conditional betas during economic expansions when the marginal utility of consumption is low.

Production frictions create cross-sectional dispersion in how firms respond to technology shocks and demand fluctuations. Firms with superior shareholder cash productivity operate closer to the efficient frontier of the production function, where diminishing returns to scale become binding [Balas et al. \(2010\)](#). These constraints generate convex adjustment costs that increase with the rate of investment, causing

high-productivity firms to bear greater systematic risk during periods of economic growth [Cooper and Priestley \(2011\)](#). The resulting variation in risk exposures across firms with different cash productivity levels creates a return premium that compensates investors for bearing productivity risk.

The link between shareholder cash productivity and expected returns strengthens through the investment channel of the stochastic discount factor. High SCP firms exhibit greater investment-cash flow sensitivity, which amplifies their exposure to aggregate investment shocks that drive the pricing kernel in production economies [Lin and Zhang \(2013\)](#). When productivity shocks hit the economy, these firms experience larger fluctuations in marginal q , leading to more volatile investment policies that covary strongly with the aggregate investment factor. This covariation generates a risk premium because investors require compensation for holding assets whose returns correlate positively with periods of high aggregate investment when consumption growth is typically high and marginal utility is low [Hou et al. \(2015\)](#).

We find strong evidence that Shareholder Cash Productivity predicts cross-sectional stock returns. A value-weighted long-short strategy that buys high SCP stocks and sells low SCP stocks generates an average monthly return of 47 basis points with a t -statistic of 3.80, yielding an annualized Sharpe ratio of 0.63. The strategy's alpha remains economically and statistically significant across all standard factor models, ranging from 36 to 49 basis points per month with t -statistics exceeding 3.47. The alpha is most conservative relative to the Fama-French five-factor model at 36 basis points monthly, consistent with the strategy's significant loading of 0.32 on the robust-minus-weak profitability factor.

The SCP premium exhibits remarkable robustness across different portfolio construction methodologies and remains substantial after accounting for transaction costs. Net of trading costs, the value-weighted NYSE-breakpoint strategy delivers 45 basis points per month with a t -statistic of 3.62, while the corresponding general-

ized alpha relative to the six-factor model equals 37 basis points with a t-statistic of 3.53. Among the largest quintile of stocks by market capitalization, the SCP strategy generates 46 basis points monthly with a t-statistic of 2.93, demonstrating that the anomaly is not confined to small, illiquid stocks where arbitrage costs might prevent price correction.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the SCP strategy maintains an alpha of 23 basis points per month with a t-statistic of 2.21. The strategy’s gross Sharpe ratio of 0.63 exceeds 96% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.60 surpasses 99% of existing strategies after transaction costs. These results establish SCP as one of the strongest cross-sectional return predictors, with a dollar invested in the strategy growing to \$6.05 gross and \$5.41 net over our sample period, ranking in the top percentile across all documented anomalies.

This paper contributes to the production-based asset pricing literature by documenting a novel link between cash flow productivity and expected returns that existing factors fail to explain. While [Novy-Marx \(2013\)](#) demonstrates that gross profitability predicts returns, and [Fama and French \(2015\)](#) incorporate operating profitability into their five-factor model, our SCP measure captures a distinct dimension of firm productivity related to the efficiency of converting operations into shareholder distributions. The SCP premium remains significant after controlling for these profitability measures, suggesting that the market prices exposure to cash productivity shocks separately from traditional profitability risks.

Our findings extend the investment-based asset pricing framework of [Hou et al. \(2015\)](#) by identifying shareholder cash productivity as a characteristic that proxies for firms’ exposure to systematic investment shocks. Unlike the investment factor constructed from asset growth or capital expenditures, SCP directly measures the productivity of capital in generating distributable cash flows—a more primitive de-

terminant of firm value in production economies. The strategy’s significant loading on the RMW factor combined with its incremental alpha indicates that SCP captures variation in conditional risk premia not spanned by existing production-based factors.

The economic magnitude and statistical robustness of the SCP premium have important implications for understanding the sources of systematic risk in production economies. The persistence of the anomaly across size groups and after transaction costs suggests that it reflects compensation for bearing fundamental macroeconomic risks rather than market frictions or limits to arbitrage. By linking cross-sectional return predictability to a theoretically motivated measure of production efficiency, we provide new evidence supporting models where asset prices emerge endogenously from firms’ optimal production and investment decisions under uncertainty [Balas et al. \(2010\)](#); [Lin and Zhang \(2013\)](#).

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S.-listed firms and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the asset pricing literature. signal construction requires two key variables from COMPUSTAT. Operating activities net cash flow (OANCF) represents the cash generated from a firm’s core business operations during the fiscal year, calculated as cash receipts from customers less cash payments to suppliers, employees, and other operating expenses. This measure reflects the actual cash flows from operating activities as reported in the statement of cash flows. Common stock (CSTK) represents the par or stated

value of common shares outstanding as reported on the balance sheet, reflecting the nominal equity capital contributed by shareholders. We construct the Shareholder Cash Productivity signal by dividing operating activities net cash flow (OANCF) by common stock (CSTK) for each firm-year observation. This ratio captures the efficiency with which firms generate operating cash flows relative to the nominal equity capital base, providing a measure of cash flow productivity per dollar of stated common equity. We calculate the signal using fiscal year-end values and require non-missing values for both OANCF and CSTK. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative common stock values are excluded from the analysis as the ratio becomes undefined or economically uninterpretable in these cases.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the SCP signal. Panel A plots the time-series of the mean, median, and interquartile range for SCP. On average, the cross-sectional mean (median) SCP is 993.99 (17.46) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input SCP data. The signal's interquartile range spans -469.69 to 493.50. Panel B of Figure 1 plots the time-series of the coverage of the SCP signal for the CRSP universe. On average, the SCP signal is available for 7.23% of CRSP names, which on average make up 7.70% of total market capitalization.

4 Does SCP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SCP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short

portfolio that buys the high SCP portfolio and sells the low SCP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short SCP strategy earns an average return of 0.47% per month with a t-statistic of 3.80. The annualized Sharpe ratio of the strategy is 0.63. The alphas range from 0.36% to 0.49% per month and have t-statistics exceeding 3.47 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.32, with a t-statistic of 6.65 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 503 stocks and an average market capitalization of at least \$1,781 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed

from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 45 bps/month with a t-statistics of 2.77. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 30-64bps/month. The lowest return, (30 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.81. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SCP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the SCP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SCP, as well as average returns and alphas for long/short trading SCP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the SCP strategy achieves an average return of 46 bps/month with a t-statistic of 2.93. Among these large cap stocks, the alphas for the SCP strat-

egy relative to the five most common factor models range from 29 to 49 bps/month with t-statistics between 1.94 and 3.68.

5 How does SCP perform relative to the zoo?

Figure 2 puts the performance of SCP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the SCP strategy falls in the distribution. The SCP strategy’s gross (net) Sharpe ratio of 0.63 (0.60) is greater than 96% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the SCP strategy (red line).² Ignoring trading costs, a \$1 invested in the SCP strategy would have yielded \$6.05 which ranks the SCP strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the SCP strategy would have yielded \$5.41 which ranks the SCP strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the SCP relative to those. Panel A shows that the SCP strategy gross alphas fall between the 75 and 90 percentiles across the five

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The SCP strategy has a positive net generalized alpha for five out of the five factor models. In these cases SCP ranks between the 90 and 97 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does SCP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of SCP with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SCP or at least to weaken the power SCP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of SCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SCP}SCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

on α from spanning tests of the form: $r_{SCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SCP. Stocks are finally grouped into five SCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SCP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SCP signal in these Fama-MacBeth regressions exceed 0.54, with the minimum t-statistic occurring when controlling for Cash-based operating profitability. Controlling for all six closely related anomalies, the t-statistic on SCP is 1.27.

Similarly, Table 5 reports results from spanning tests that regress returns to the SCP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SCP strategy earns alphas that range from 28-38bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.71, which is achieved when controlling for Cash-based operating profitability. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SCP trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.21.

7 Does SCP add relative to the whole zoo?

Finally, we can ask how much adding SCP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the SCP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes SCP grows to \$32.50.

8 Conclusion

This study provides compelling evidence that Shareholder Cash Productivity (SCP) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that SCP-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related profitability measures.

The key findings reveal that a value-weighted long/short strategy based on SCP delivers impressive performance metrics, with annualized Sharpe ratios of 0.63 (gross)

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which SCP is available.

and 0.60 (net), indicating strong risk-adjusted returns that remain economically meaningful after accounting for transaction costs. Most notably, the strategy generates statistically significant abnormal returns of 38 basis points per month (t-statistic = 3.62) relative to the Fama-French five-factor model augmented with momentum, suggesting that SCP captures unique information about firm fundamentals not explained by existing factor models.

Perhaps most compelling is the signal’s resilience when tested against its closest competitors in the factor zoo. Even after controlling for six highly related strategies—including cash-based operating profitability, RD-adjusted operating profitability, and various liquidity and profitability measures—SCP maintains a significant alpha of 23 basis points per month (t-statistic = 2.21). This finding underscores that SCP provides incremental predictive power beyond existing profitability and cash flow metrics, suggesting it captures a distinct dimension of firm quality related to how efficiently companies convert shareholder capital into cash flows.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, SCP represents a meaningful addition to the growing literature on profitability-based anomalies, offering new insights into the relationship between cash flow generation efficiency and expected returns. For practitioners, the signal’s strong performance after transaction costs and its orthogonality to existing factors suggest it could serve as a valuable component in quantitative investment strategies and multi-factor models.

However, several limitations warrant consideration. First, while our analysis follows best practices for anomaly testing, the true out-of-sample performance of SCP remains to be seen as the signal becomes more widely known and potentially arbitrated away. Second, our study focuses primarily on U.S. equity markets, and the international robustness of SCP requires further investigation. Third, while we control for numerous related factors, the economic mechanism driving the SCP

premium deserves deeper theoretical and empirical exploration.

Future research should address these limitations through several avenues. International studies could examine whether SCP predicts returns in developed and emerging markets, providing insights into the global applicability of the signal. Time-series analysis of the SCP premium could reveal whether its predictive power varies with macroeconomic conditions or market regimes. Additionally, investigating the micro-foundations of why high SCP firms earn higher returns—whether through risk-based or behavioral channels—would enhance our understanding of this anomaly. Finally, examining the interaction between SCP and corporate actions such as capital allocation decisions, MA activity, and dividend policy could provide valuable insights into the real economic implications of shareholder cash productivity.

In conclusion, Shareholder Cash Productivity emerges as a robust and economically significant predictor of stock returns that withstands rigorous empirical scrutiny. Its ability to generate alpha beyond existing factors and maintain significance after transaction costs positions it as a valuable addition to the asset pricing literature and a potentially useful tool for investment management.

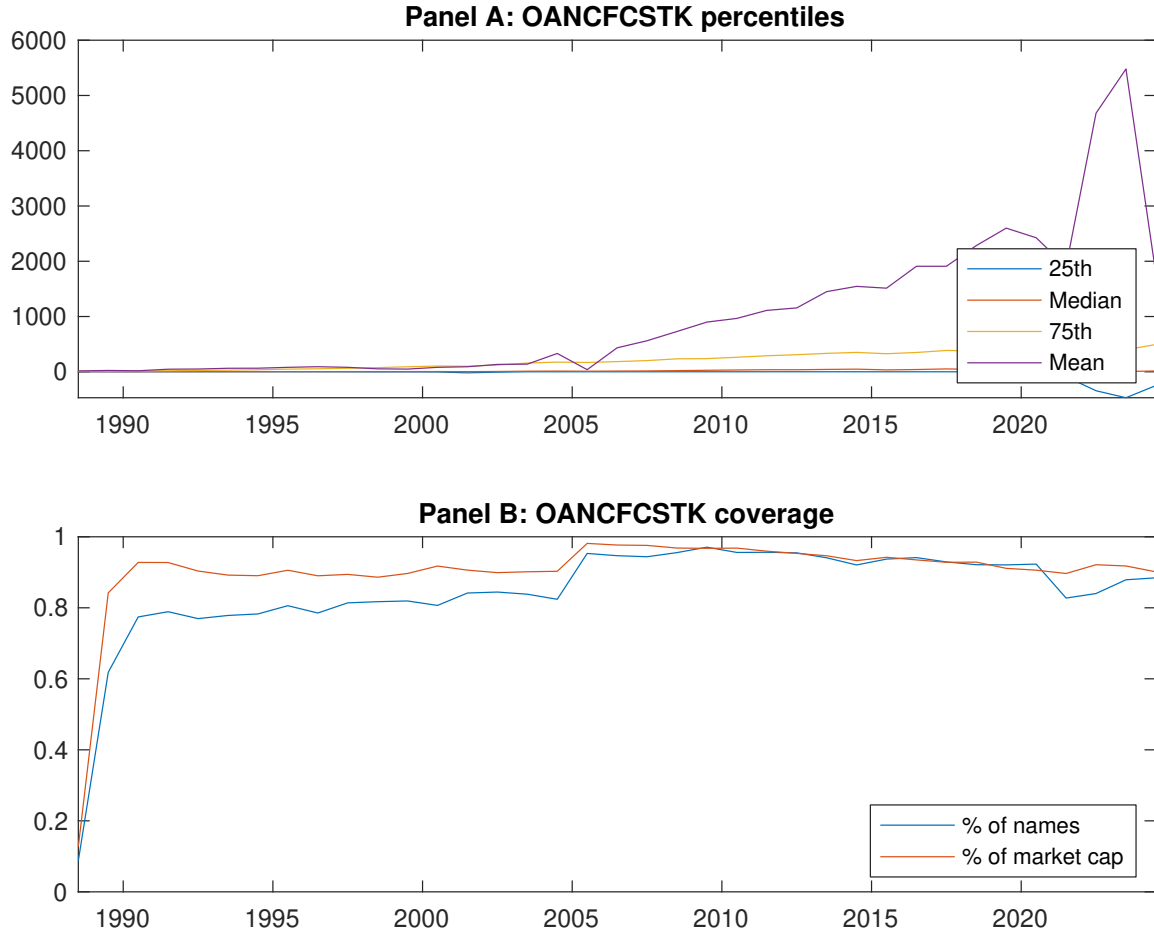


Figure 1: Times series of SCP percentiles and coverage.
This figure plots descriptive statistics for SCP. Panel A shows cross-sectional percentiles of SCP over the sample. Panel B plots the monthly coverage of SCP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SCP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on SCP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.49 [1.97]	0.64 [3.58]	0.71 [3.52]	0.70 [3.05]	0.96 [3.81]	0.47 [3.80]
α_{CAPM}	-0.32 [-3.47]	0.07 [0.91]	0.03 [0.47]	-0.09 [-1.57]	0.11 [1.42]	0.43 [3.47]
α_{FF3}	-0.30 [-3.96]	0.02 [0.26]	-0.01 [-0.27]	-0.09 [-1.52]	0.16 [2.50]	0.47 [4.36]
α_{FF4}	-0.30 [-3.88]	-0.01 [-0.09]	-0.01 [-0.13]	-0.05 [-0.85]	0.19 [2.83]	0.49 [4.50]
α_{FF5}	-0.10 [-1.46]	-0.19 [-3.05]	-0.09 [-1.68]	-0.02 [-0.40]	0.27 [4.08]	0.36 [3.48]
α_{FF6}	-0.11 [-1.56]	-0.20 [-3.10]	-0.08 [-1.47]	0.01 [0.10]	0.28 [4.20]	0.38 [3.62]
Panel B: Fama and French (2018) 6-factor model loadings for SCP-sorted portfolios						
β_{MKT}	0.99 [58.13]	0.89 [56.39]	0.97 [73.34]	1.03 [72.67]	1.07 [65.35]	0.08 [3.15]
β_{SMB}	0.23 [9.42]	-0.03 [-1.48]	-0.02 [-1.17]	-0.02 [-1.09]	0.00 [0.09]	-0.23 [-6.02]
β_{HML}	0.16 [5.37]	0.03 [1.14]	0.13 [5.44]	0.02 [0.96]	-0.14 [-5.05]	-0.30 [-6.60]
β_{RMW}	-0.41 [-13.31]	0.28 [9.91]	0.13 [5.30]	-0.10 [-4.02]	-0.09 [-3.11]	0.32 [6.65]
β_{CMA}	-0.12 [-2.68]	0.37 [9.08]	0.09 [2.72]	-0.05 [-1.40]	-0.26 [-6.15]	-0.14 [-2.09]
β_{UMD}	0.01 [0.82]	0.01 [0.63]	-0.02 [-1.36]	-0.04 [-3.55]	-0.02 [-1.14]	-0.03 [-1.23]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1474	503	660	668	690	
me (\$10 ⁶)	1781	3349	2803	2759	4874	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the SCP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.47 [3.80]	0.43 [3.47]	0.47 [4.36]	0.49 [4.50]	0.36 [3.48]	0.38 [3.62]
Quintile	NYSE	EW	0.45 [2.77]	0.50 [3.00]	0.40 [2.81]	0.33 [2.32]	0.06 [0.50]	0.01 [0.11]
Quintile	Name	VW	0.48 [2.33]	0.75 [3.82]	0.68 [4.07]	0.63 [3.75]	0.27 [1.80]	0.25 [1.64]
Quintile	Cap	VW	0.47 [3.35]	0.34 [2.46]	0.43 [3.73]	0.46 [4.01]	0.50 [4.36]	0.52 [4.49]
Decile	NYSE	VW	0.68 [3.68]	0.82 [4.49]	0.80 [5.08]	0.78 [4.86]	0.55 [3.63]	0.54 [3.52]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.45 [3.62]	0.42 [3.40]	0.44 [4.11]	0.46 [4.23]	0.36 [3.45]	0.37 [3.53]
Quintile	NYSE	EW	0.30 [1.81]	0.35 [2.01]	0.25 [1.72]	0.22 [1.47]		
Quintile	Name	VW	0.45 [2.16]	0.73 [3.74]	0.65 [3.91]	0.62 [3.72]	0.30 [2.00]	0.28 [1.87]
Quintile	Cap	VW	0.46 [3.24]	0.33 [2.37]	0.40 [3.50]	0.42 [3.69]	0.47 [4.16]	0.49 [4.21]
Decile	NYSE	VW	0.64 [3.48]	0.80 [4.40]	0.77 [4.85]	0.76 [4.73]	0.56 [3.67]	0.55 [3.60]

Table 3: Conditional sort on size and SCP

This table presents results for conditional double sorts on size and SCP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SCP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SCP and short stocks with low SCP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	SCP Quintiles					SCP Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.25 [0.58]	0.30 [0.80]	0.74 [2.71]	0.89 [3.25]	1.01 [3.29]	0.76 [2.85]	0.98 [3.74]	0.79 [3.65]	0.75 [3.41]	0.32 [1.63]	0.30 [1.52]
	(2)	0.33 [0.82]	0.56 [2.09]	0.92 [3.42]	0.92 [3.27]	0.90 [3.19]	0.57 [2.58]	0.81 [3.79]	0.65 [3.76]	0.63 [3.59]	0.22 [1.51]	0.22 [1.46]
	(3)	0.35 [1.11]	0.69 [2.82]	0.83 [3.02]	0.87 [3.27]	0.97 [3.50]	0.62 [3.77]	0.72 [4.38]	0.65 [4.16]	0.59 [3.73]	0.30 [2.10]	0.26 [1.82]
	(4)	0.56 [2.08]	0.73 [3.15]	0.81 [3.35]	0.80 [3.23]	1.08 [4.07]	0.51 [4.16]	0.52 [4.11]	0.49 [3.96]	0.46 [3.63]	0.28 [2.36]	0.26 [2.14]
	(5)	0.51 [2.37]	0.70 [3.85]	0.66 [3.31]	0.67 [3.03]	0.97 [3.66]	0.46 [2.93]	0.29 [1.94]	0.39 [3.03]	0.43 [3.24]	0.47 [3.60]	0.49 [3.68]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	SCP Quintiles					SCP Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	434	438	448	447	446	45	39	50	55	68	
	(2)	130	132	131	131	130	86	90	93	94	94	
	(3)	88	89	89	88	88	150	159	155	159	158	
	(4)	73	73	73	73	73	325	333	327	329	333	
(5)	64	65	64	64	64	1946	2518	2337	2017	3606		

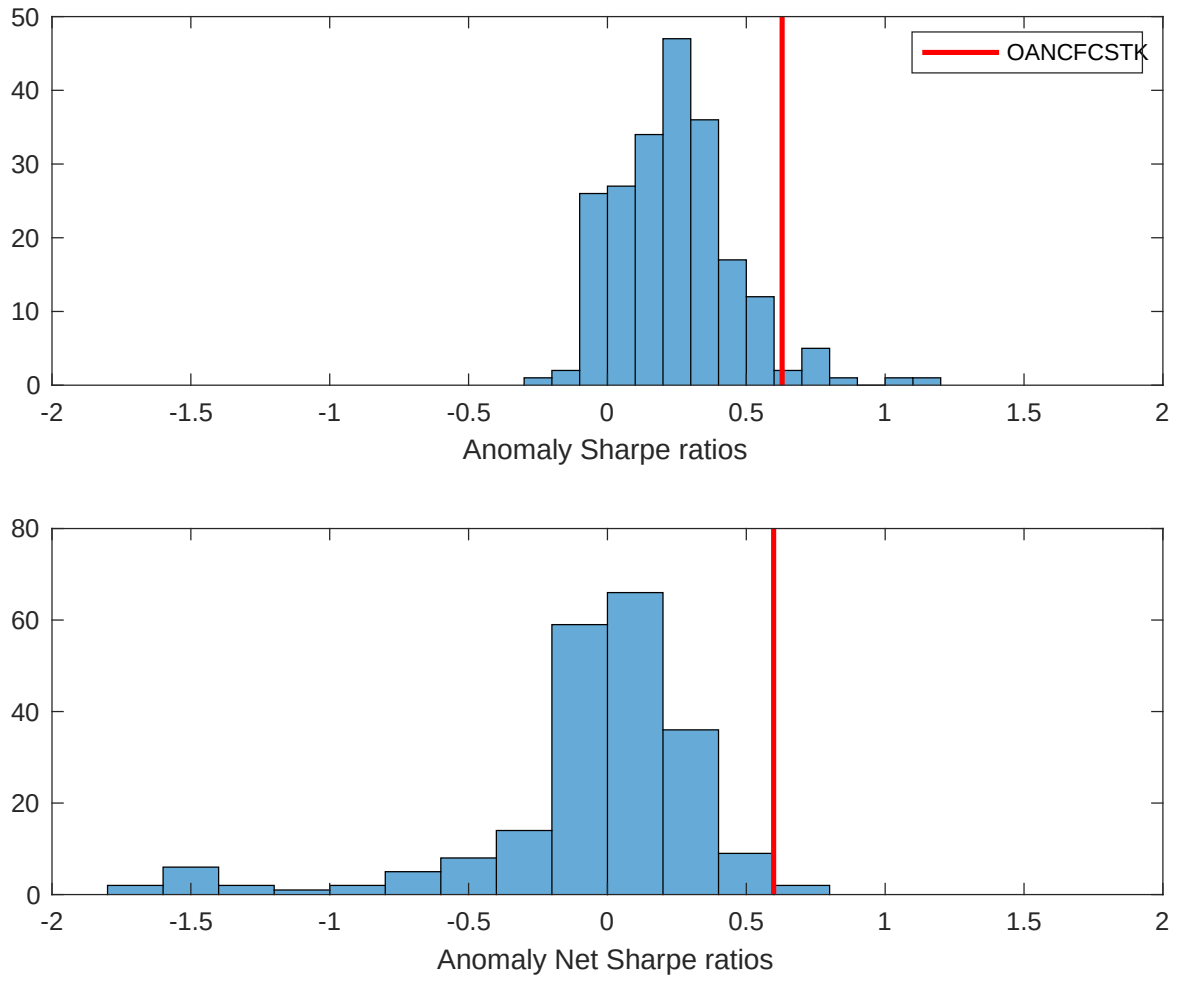


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SCP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

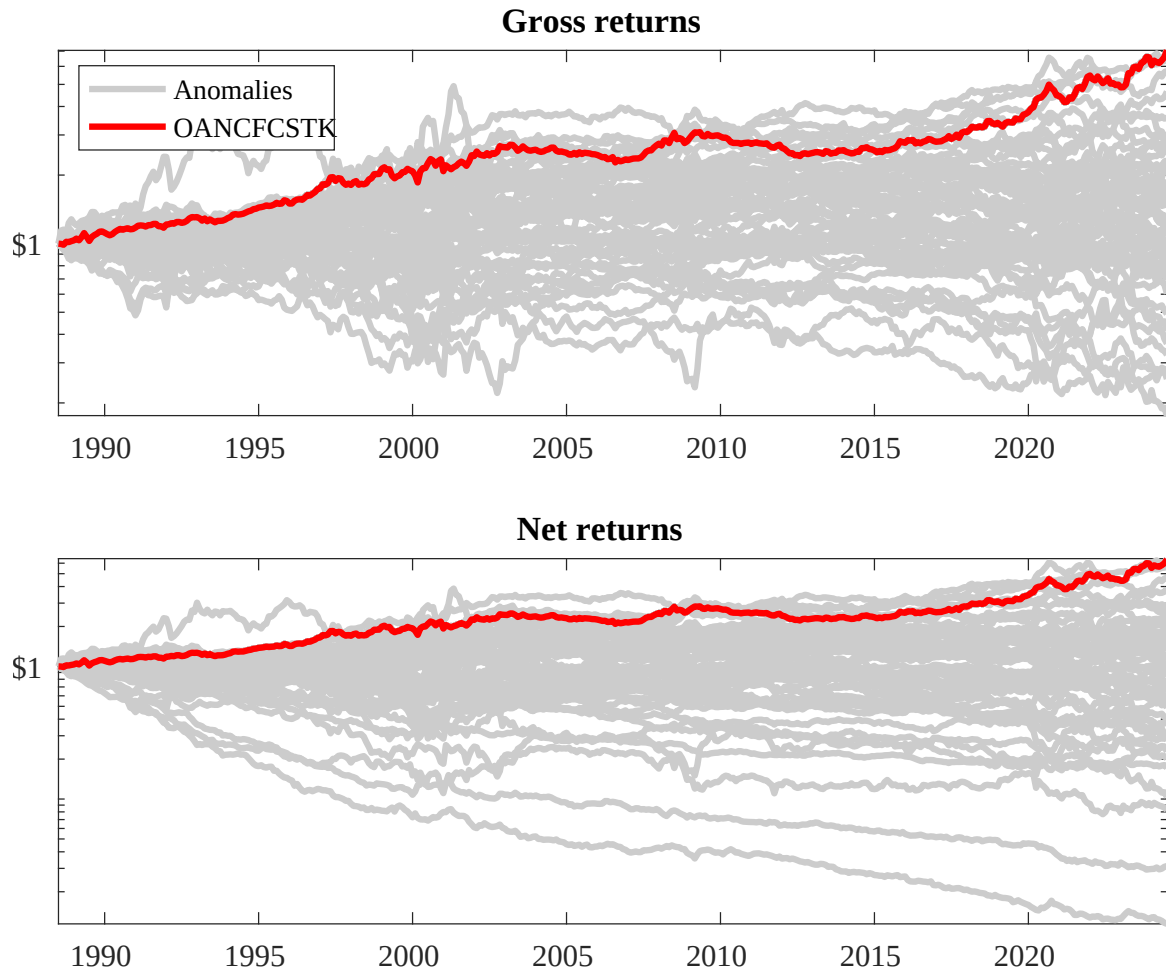
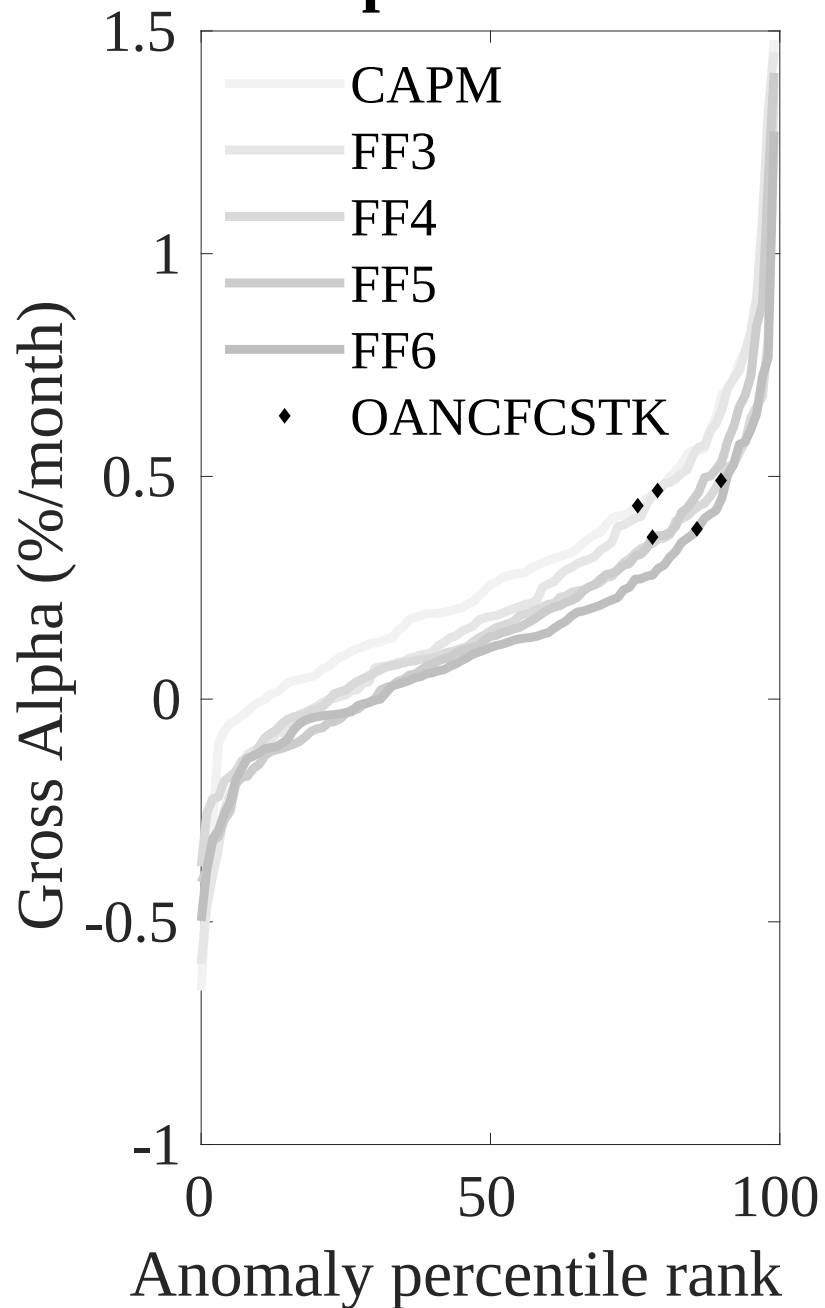


Figure 3: Dollar invested.

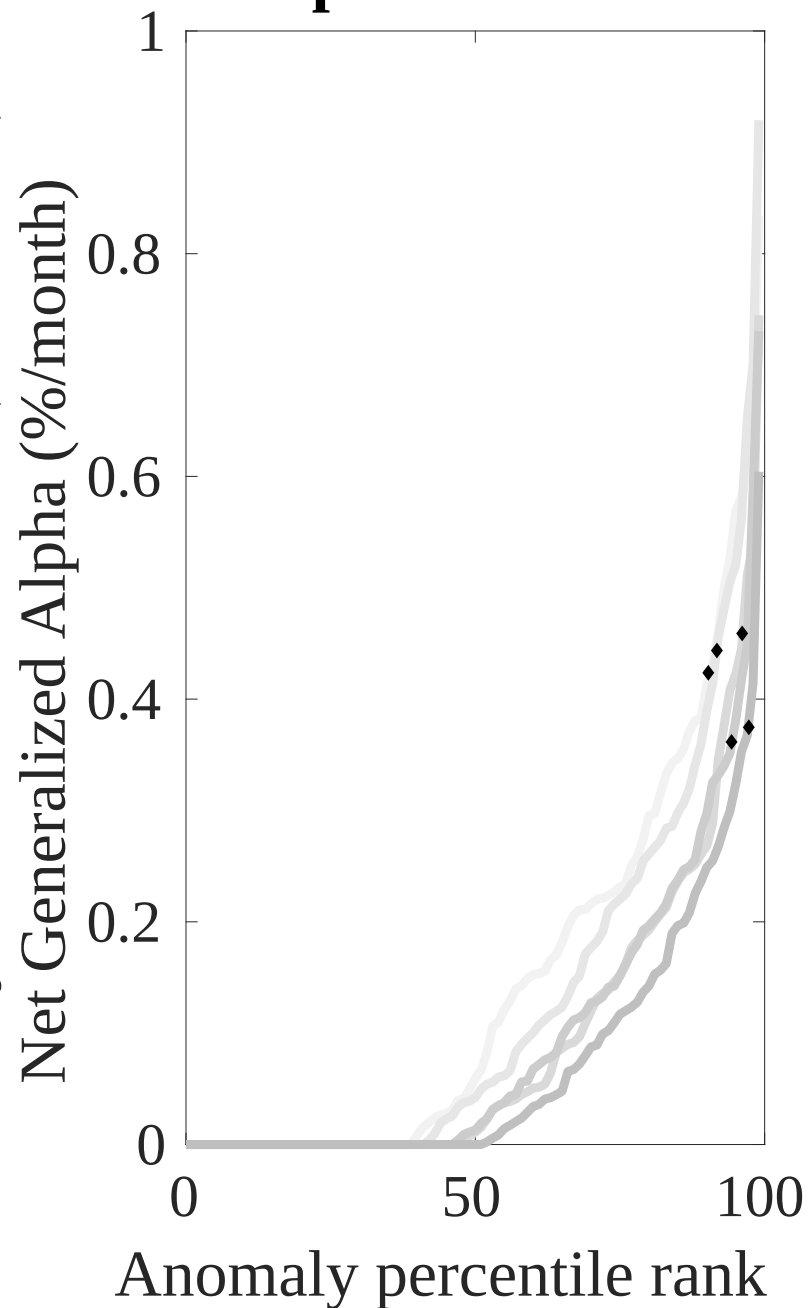
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SCP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



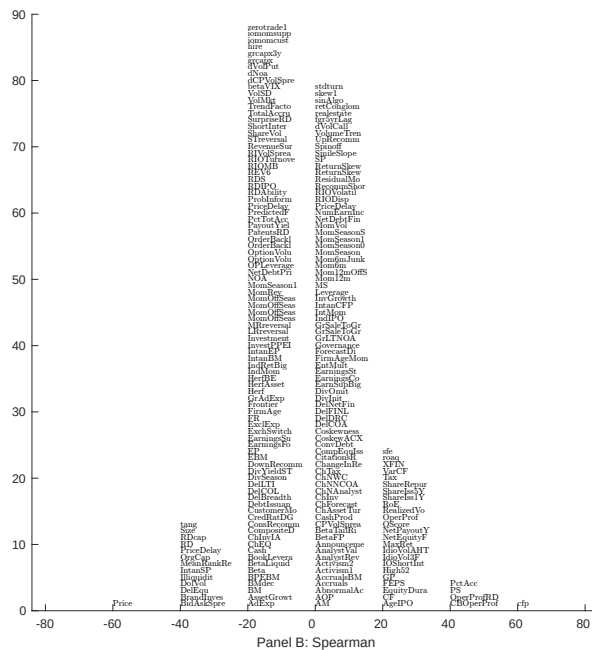
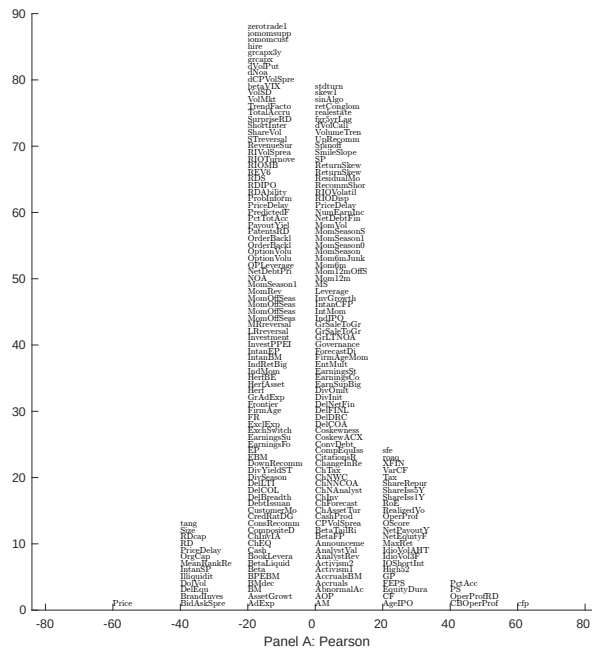
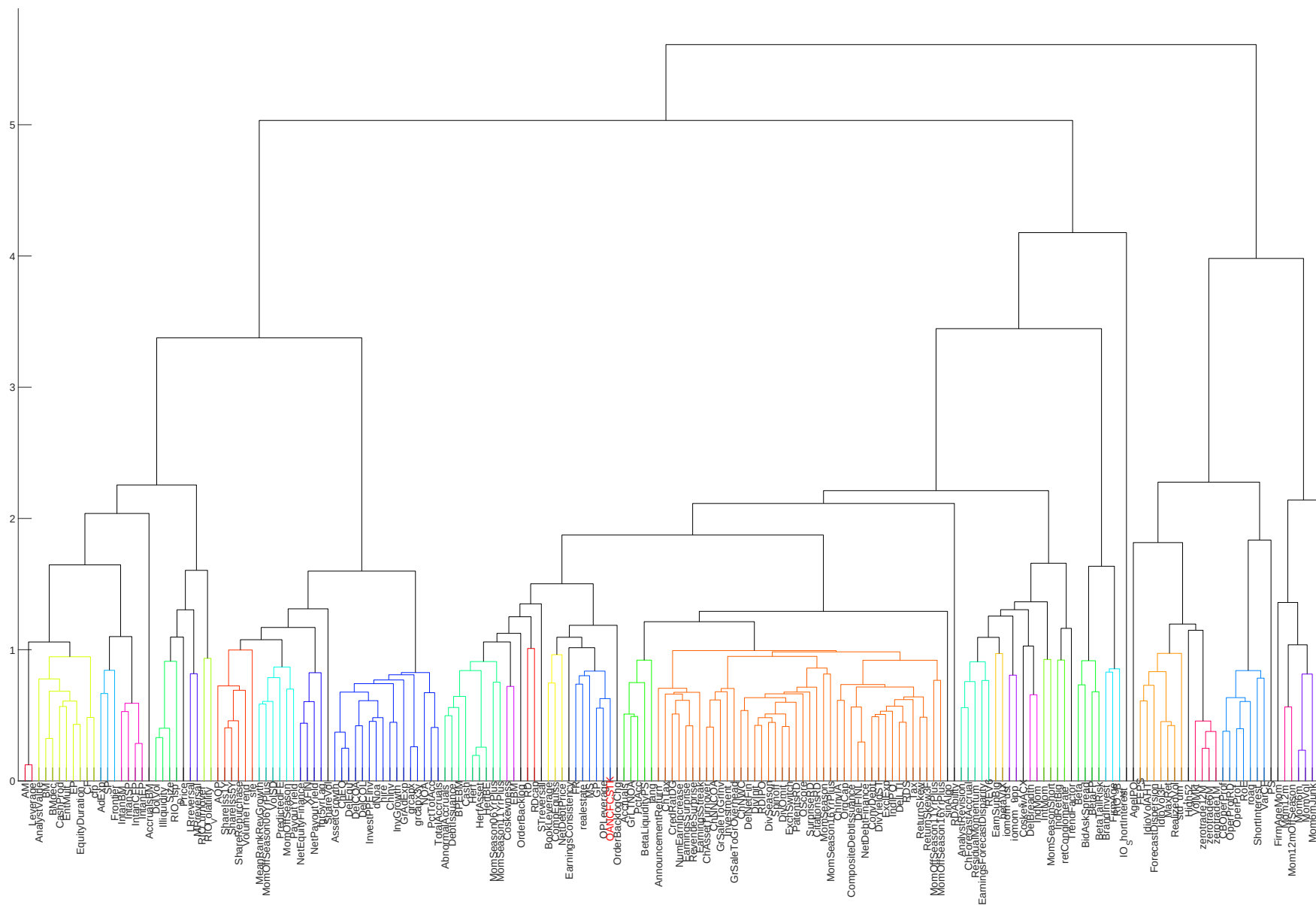


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with SCP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



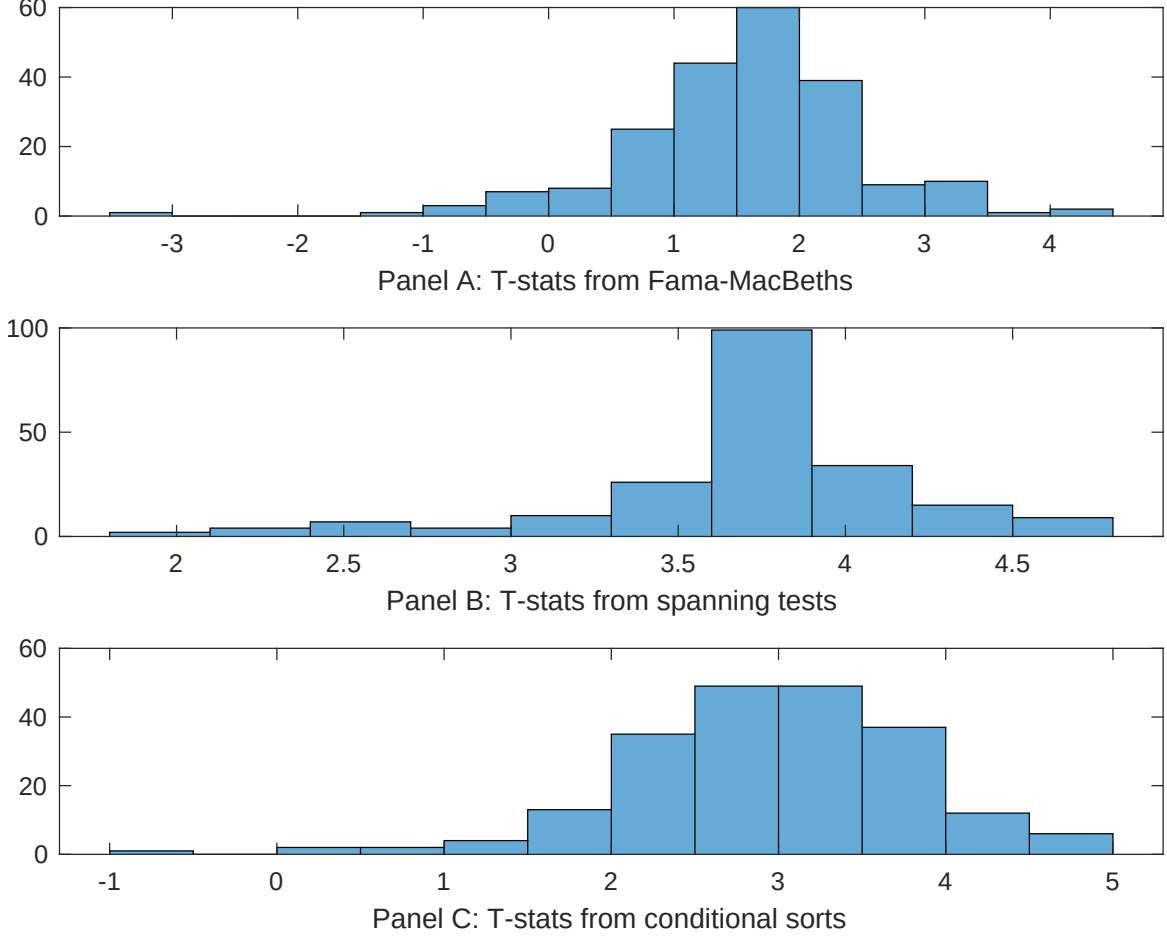


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of SCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SCP}SCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SCP. Stocks are finally grouped into five SCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on SCP, and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{SCP}SCP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Cash-based operating profitability, Operating profitability RD adjusted, Amihud's illiquidity, gross profits / total assets, Return on assets (qtrly), Past trading volume. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.96 [3.00]	0.90 [2.71]	0.92 [3.16]	0.82 [2.51]	0.11 [4.04]	0.12 [3.78]	0.98 [3.15]
SCP	0.56 [0.54]	0.20 [1.76]	0.30 [2.59]	0.18 [1.44]	0.82 [0.92]	0.30 [2.69]	0.12 [1.27]
Anomaly 1	0.18 [4.24]						0.15 [4.08]
Anomaly 2		0.17 [3.38]					-0.29 [-0.61]
Anomaly 3			0.36 [2.56]				0.55 [2.03]
Anomaly 4				0.72 [3.79]			0.43 [1.87]
Anomaly 5					0.41 [2.57]		0.45 [3.09]
Anomaly 6						0.74 [2.48]	0.76 [2.79]
# months	427	427	427	432	427	427	427
$\bar{R}^2(\%)$	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the SCP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{SCP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Cash-based operating profitability, Operating profitability RD adjusted, Amihud's illiquidity, gross profits / total assets, Return on assets (qtrly), Past trading volume. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.28 [2.71]	0.29 [2.78]	0.38 [3.58]	0.31 [2.98]	0.35 [3.31]	0.38 [3.52]	0.23 [2.21]
Anomaly 1	27.72 [6.15]						20.42 [2.98]
Anomaly 2		19.61 [4.55]					-4.37 [-0.70]
Anomaly 3			-8.91 [-1.08]				-26.82 [-2.20]
Anomaly 4				25.75 [6.06]			16.20 [3.18]
Anomaly 5					17.97 [3.86]		12.18 [2.51]
Anomaly 6						3.26 [0.44]	32.66 [3.01]
mkt	10.73 [4.18]	11.57 [4.32]	7.35 [2.59]	7.66 [3.02]	11.00 [4.10]	9.20 [2.93]	14.83 [4.61]
smb	-14.15 [-3.72]	-15.60 [-4.01]	-11.14 [-1.14]	-22.54 [-6.18]	-17.24 [-4.47]	-23.78 [-3.11]	-15.84 [-1.70]
hml	-18.14 [-3.79]	-20.93 [-4.30]	-27.62 [-5.75]	-18.68 [-3.91]	-23.06 [-4.79]	-29.80 [-5.95]	-16.59 [-3.28]
rmw	17.89 [3.49]	17.36 [3.06]	29.86 [5.81]	18.25 [3.56]	15.54 [2.45]	31.86 [6.63]	-0.94 [-0.14]
cma	-19.99 [-3.06]	-15.73 [-2.38]	-14.05 [-2.08]	-10.32 [-1.58]	-15.69 [-2.36]	-14.80 [-2.19]	-15.74 [-2.39]
umd	-4.92 [-2.15]	-5.17 [-2.19]	-2.88 [-1.22]	-3.64 [-1.61]	-7.20 [-2.77]	-2.13 [-0.83]	-3.69 [-1.34]
# months	428	428	428	432	428	428	428
$\bar{R}^2(\%)$	39	37	34	40	36	34	42

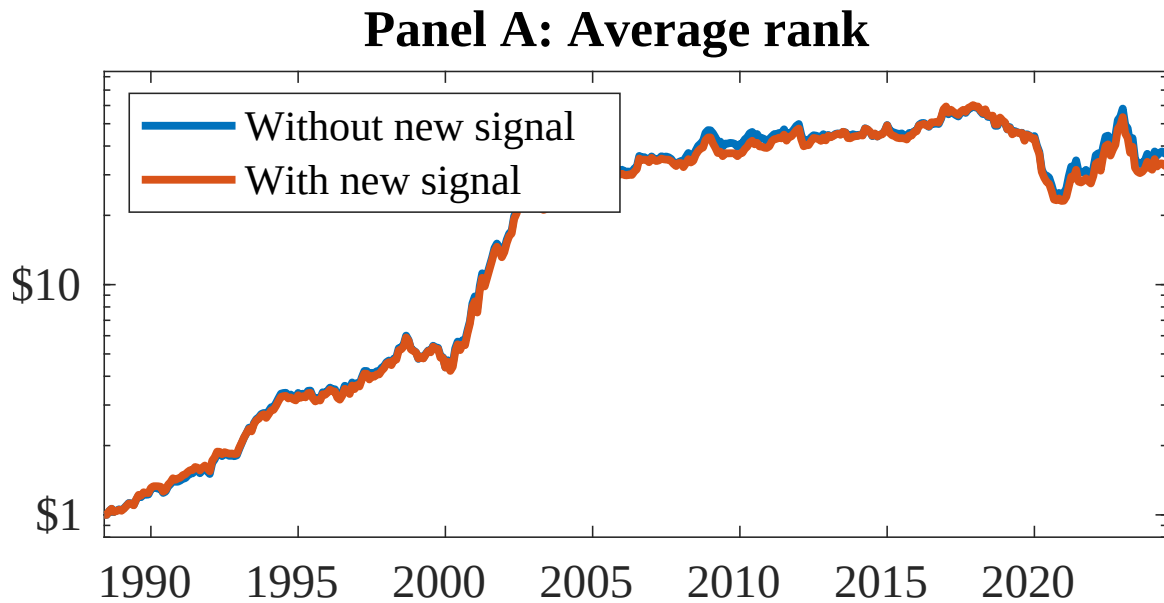


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as SCP. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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